SUPERIOR UNIVERSITY, LAHORE

FACULTY OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

COURSE PROJECT • GENERATIVE ARTIFICIAL INTELLIGENCE (BSAI 5TH SEM)

NEON JARVIS AI

An Autonomous Holographic Personal AI Operating System Featuring Multilingual Voice Interaction (Urdu, English, Punjabi), Grok-2 LLM Integration, Real-Time Web Intelligence, and YouTube Voice Control.

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COURSE & REPOSITORY DETAILS

Generative AI — 5th Semester

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01 Executive Summary & Abstract

NEON JARVIS AI is a full-featured, futuristic web-based personal AI assistant and personal operating system inspired by Iron Man's iconic J.A.R.V.I.S. companion. Engineered on Next.js 15 App Router, TypeScript, and TailwindCSS, the application synthesizes high-fidelity voice computing, multilingual intelligence (English, Urdu, and Pakistani Punjabi), real-time web querying, and autonomous media control within a cinematic, sci-fi holographic user interface.

Unlike conventional single-turn conversational chatbots that produce generic text streams, NEON JARVIS operates as an active desktop companion. It continuously listens for wake words (*"Jarvis"*, *"Assistant"*), displays an audio-reactive cybernetic robotic AI face across four emotional states, and instantly executes complex real-world voice tasks such as querying live global weather, fetching real-time web intelligence, and streaming YouTube videos while responding to immediate cessation commands across regional languages (*"Stop"*, *"Ruk jao"*, *"Bus karo"*, *"Band karo"*).

02 Problem Statement & Motivation

Most commercial AI conversational interfaces suffer from three systemic constraints:

- **Monolingual & Standardized Bias:** Existing systems cater predominantly to standard English, failing to recognize regional colloquial dialects such as Roman Urdu and Pakistani Punjabi naturally mixed in daily speech.
- **Passive Text-Only Modality:** Standard chatbots lack physical or animated presence, failing to communicate state transitions (listening, computing, verbalizing) intuitively.
- **Action Isolation:** Conventional LLM chatbots generate theoretical advice rather than directly controlling client-side actuators (such as opening, seeking, and halting media streams).

NEON JARVIS resolves these limitations by creating a tight integration between natural language understanding, real-time API routes, Web Speech / Web Audio APIs, and an expressive visual AI core.

03 Architectural Blueprint & Tech Stack

The architecture follows a modular, edge-compatible Next.js 15 design structured into clean decoupled layers: presentation components, state management, orchestration services, and resilient API handlers.

Layer	Technology	Responsibility
Framework	Next.js 15 (App Router)	Server-side rendering, production bundling, optimized asset loading.
Language	TypeScript 5.7	Strict interface contracts for telemetry, intents, chat, and media models.
Styling & FX	TailwindCSS + Framer Motion	Custom neon color tokens, glassmorphism, scanlines, particle physics.
State Core	Zustand (Store)	Global reactive state with localStorage persistence for user settings.
Primary LLM	Grok-2 API (xAI)	High-reasoning conversational intelligence with multilingual system prompt.
Speech Synthesis	ElevenLabs + Web Speech	Multilingual neural audio stream with browser TTS automatic fallback.
Web Audio Engine	Web Audio API	Zero-dependency procedural sound generator (wake chime, reactor hum).
Media Engine	YouTube Subsystem	Embedded player with autonomous scraping resolver and voice control.

SYSTEM RELIABILITY PRINCIPLE: THE AUTONOMOUS FALLBACK ENGINE

A core requirement of this project was resilience. If external API keys (Grok, ElevenLabs, Google Search) are missing or rate-limited, NEON JARVIS seamlessly switches to local neural heuristics: the DuckDuckGo live intelligence engine, Open-Meteo live weather, native Web Speech synthesis, and YouTube

search scrapers. The platform is guaranteed to operate smoothly out-of-the-box.

04 Multilingual Speech & Purity Pipeline

NEON JARVIS handles code-switching between English, Urdu, and Pakistani Punjabi. The system uses a specialized intent parser that extracts linguistic markers and maps the user's intent:

- **Punjabi Detection:** Identifies phonetic tokens such as *"vich"*, *"kivein"*, *"ki haal"*, *"tussi"*, *"gal karo"*, *"changa"*, *"dasso"*, and responds in warm, witty Punjabi.
- **Urdu Detection:** Matches vocabulary like *"kaise ho"*, *"kya chal raha hai"*, *"mausam"*, *"batao"*, *"shukriya"*, and replies respectfully with polite Pakistani assistant phrasing (*"Jee Janab, hukum karein"*).
- **English Detection:** Replies in classic British/crisp Jarvis persona (*"Right away, Sir"*, *"All systems operational"*).

SPEECH PURITY STANDARD (ELIMINATION OF MARKDOWN & URL ARTIFACTS)

A common pitfall of LLM voice agents is pronouncing raw URLs, formatting asterisks, and hash symbols (e.g. reading out *"asterisk asterisk bold asterisk asterisk"* or *"h-t-t-p-s colon slash slash"*). NEON JARVIS implements a strict sanitizer:

```
// utils/cleanSpeechText.ts
export function cleanSpeechText(text: string): string {
  return text
    .replace(/```[\s\S]*?```/g, "")      // Code blocks
    .replace(/`([^`]+)`/g, "$1")         // Inline code
    .replace(/\[([^\]]+)\]\([^)]+\)/g, "$1") // Markdown links
    .replace(/https?:\/\/\S+/gi, "")     // Raw URLs
    .replace(/^#+\s+/gm, "")             // Headers
    .replace(/[*_~]{1,3}/g, "")          // Bold / Italics
    .replace(/\s{2,}/g, " ")             // Extra spaces
    .trim();
}
```

05 Voice Directive Engine & Instant Stop Directives

The voice pipeline allows immediate hands-free interaction. When the user speaks, speech recognition matches patterns to actionable commands:

Directive Type	Voice Triggers	Automated Action
Media Playback	*"Play [song/naat/artist]"*, *"Coke Studio chalao"*, *"Atif Aslam ke gane sunao"*	Extracts query, queries YouTube resolver, launches HUD player with autoplay.
Instant Cessation	*"Stop"*, *"Ruk jao"*, *"Bus karo"*, *"Band karo"*, *"Roko"*	Halts YouTube iframe immediately, cancels active speech synthesis, sounds stop chirp.
Live Weather	*"Kal ka weather batao"*, *"Lahore weather report"*	Queries Open-Meteo geocoding & forecast API, displays weather card, speaks temperature in °C.
Real-Time Search	*"Search latest AI news"*, *"Dhoondo SpaceX"*	Queries live web engine, summarizes top hits, speaks natural synopsis without URLs.
Temporal Status	*"Kya waqt hua hai"*, *"What time is it"*	Announces current local time and date in the matching language.

06 Holographic AI Face & Sci-Fi Visual Synthesis

The visual centerpiece of NEON JARVIS is an animated robotic AI face engineered entirely with vector SVG, TailwindCSS, and Framer Motion:

- **Audio-Reactive Optics:** Dual cybernetic eyes that dilate and glow with variable intensity proportional to microphone input and vocal output.
- **Oscillating Mouth Soundwave:** Nine digital equalizer columns representing acoustic spectrum distribution across the lower jaw plate.
- **Bi-Directional Gyroscope:** Concentric circular reticles with degree ticks rotating continuously to evoke an Arc Reactor telemetry system.

- **Procedural Particle Mesh:** Interactive HTML5 canvas rendering floating cyan/purple nodes that dynamically link with laser filaments when in proximity.

07 Verification, Build Metrics & Delivery

The project was validated via a clean production build (`next build`) and end-to-end API route testing via HTTP protocols:

PRODUCTION COMPILATION VERIFICATION

```
✓ Compiled successfully in 10.9s
✓ Linting and checking validity of types
✓ Collecting page data
✓ Generating static pages (9/9)
✓ Finalizing page optimization

Route (app)                                 Size     First Load JS
┌ ○ /                                       63 kB    165 kB
├ ƒ /api/chat                              136 B    102 kB
├ ƒ /api/search                            136 B    102 kB
├ ƒ /api/tts                               136 B    102 kB
├ ƒ /api/weather                           136 B    102 kB
└ ƒ /api/youtube                           136 B    102 kB
```

08 Conclusion & Repository Submission

NEON JARVIS AI demonstrates the synthesis of modern web technologies, Generative AI models, and natural human-computer interaction. By uniting multilingual fluency, real-time tools, immediate voice cessation controls, and a sci-fi user experience, it establishes a benchmark for next-generation personal AI systems.

ACADEMIC SUBMISSION MANIFEST

- **Student Name:** Muhammad Maheem
- **Roll Number:** SU92-BSAIM-F24-054
- **Degree & Section:** BSAI, 5th Semester (BSAI-5A)
- **Institution:** Superior University, Lahore
- **GitHub Repository:** https://github.com/MuhammadMaheem/neon-jarvis-ai

- **Local Document Report:** `~/Documents/NEON_JARVIS_PROJECT_REPORT.pdf`
- **Submission Zip Archive:** `~/Documents/NEON_JARVIS_AI_SUBMISSION.zip`